

ArtisanHub – Handmade Products E-Commerce Platform

CHAPTER 1

1. Introduction

ArtisanHub is an online e-commerce platform designed to connect skilled artisans with customers who appreciate handmade products. The platform provides a digital marketplace where artisans can showcase and sell unique handmade items such as jewelry, pottery, paintings, candles, home décor, embroidery, and handcrafted gifts. Customers can browse products, place orders, make secure payments, track deliveries, and provide reviews. The platform also supports artisan profile management, product categorization, inventory management, and order processing. By offering a dedicated marketplace for handmade products, ArtisanHub helps preserve traditional craftsmanship while making it easier for artisans to reach customers across different locations.

1.1 Objective

The primary objective of ArtisanHub is to develop a reliable and efficient database system for managing an online marketplace dedicated to handmade products. The system aims to organize customer information, artisan details, product inventory, orders, payments, deliveries, and customer feedback in a structured manner using SQL. It ensures accurate data storage, minimizes redundancy, and supports smooth business operations.

1.2 Key Objectives

- Develop a well-structured relational database for the ArtisanHub platform.
- Maintain customer and artisan information securely.
- Store and manage product details and categories efficiently.
- Process customer orders and payment records accurately.
- Track product inventory and delivery status.
- Allow customers to submit ratings and reviews.
- Improve data consistency through normalization and database relationships.
- Support efficient querying and reporting using SQL.

1.3 Motive of the Project

Many talented artisans struggle to sell their handmade products because they have limited access to online marketplaces and digital business tools. Existing e-commerce platforms often focus on mass-produced products, making it difficult for small artisans to gain visibility. The motive behind ArtisanHub is to create a dedicated platform that promotes handmade products, supports local artisans, and provides customers with access to authentic handcrafted items. The project also demonstrates the practical application of SQL database concepts in solving real-world business problems.

1.4 Problem Statement

Small-scale artisans often face challenges in managing product information, customer orders, payments, and inventory using manual methods. These methods can lead to data duplication, errors, delayed order processing, and poor customer service. There is a need for a centralized database system that efficiently manages all business operations while ensuring data accuracy, security, and quick access to information. ArtisanHub addresses these challenges by providing a structured database solution for managing the complete e-commerce process.

1.5 Business Requirements

- Customers should be able to register and manage their accounts.
- Artisans should be able to add, update, and manage their products.
- Products should be organized into categories.
- Customers should be able to search, view, and purchase products.
- The system should manage shopping carts and orders.
- Payment details should be stored securely.
- Delivery information should be tracked.
- Customers should be able to provide ratings and reviews.
- The system should generate accurate reports for products, orders, and sales.

1.6 Functional Requirements

- Customer registration and login.
- Artisan registration and profile management.
- Product management (Add, Update, Delete, View).
- Category management.
- Shopping cart management.
- Order placement and order history.
- Payment processing and recording.
- Delivery tracking.
- Review and rating management.
- Wishlist management.
- Inventory management.
- Search and filter products.

1.7 Non-Functional Requirements

- The database should provide fast query performance.
- The system should maintain data accuracy and consistency.
- Customer information should be stored securely.
- The platform should be available whenever users access it.
- The database should support future expansion with additional features.
- The system should be reliable and minimize data loss.
- The application should have an easy-to-use interface.
- Regular data backup should be supported.

1.8 Stakeholders

The main stakeholders involved in the ArtisanHub platform are:

- Customers who purchase handmade products.
- Artisans or sellers who list and sell their products.
- System Administrator who manages the platform and database.
- Delivery Partners responsible for shipping orders.
- Payment Service Providers who process customer payments.
- Project Developers who design, develop, and maintain the system.

1.9 Conclusion

ArtisanHub is a unique e-commerce platform that supports artisans by providing a dedicated online marketplace for handmade products. The project demonstrates how SQL databases can effectively manage customers, artisans, products, orders, payments, deliveries, and reviews through a well-designed relational database. The proposed system improves efficiency, reduces manual work, ensures data integrity, and provides a better shopping experience for customers while helping artisans expand their businesses. The project also serves as a practical implementation of database management concepts in a real-world e-commerce environment.